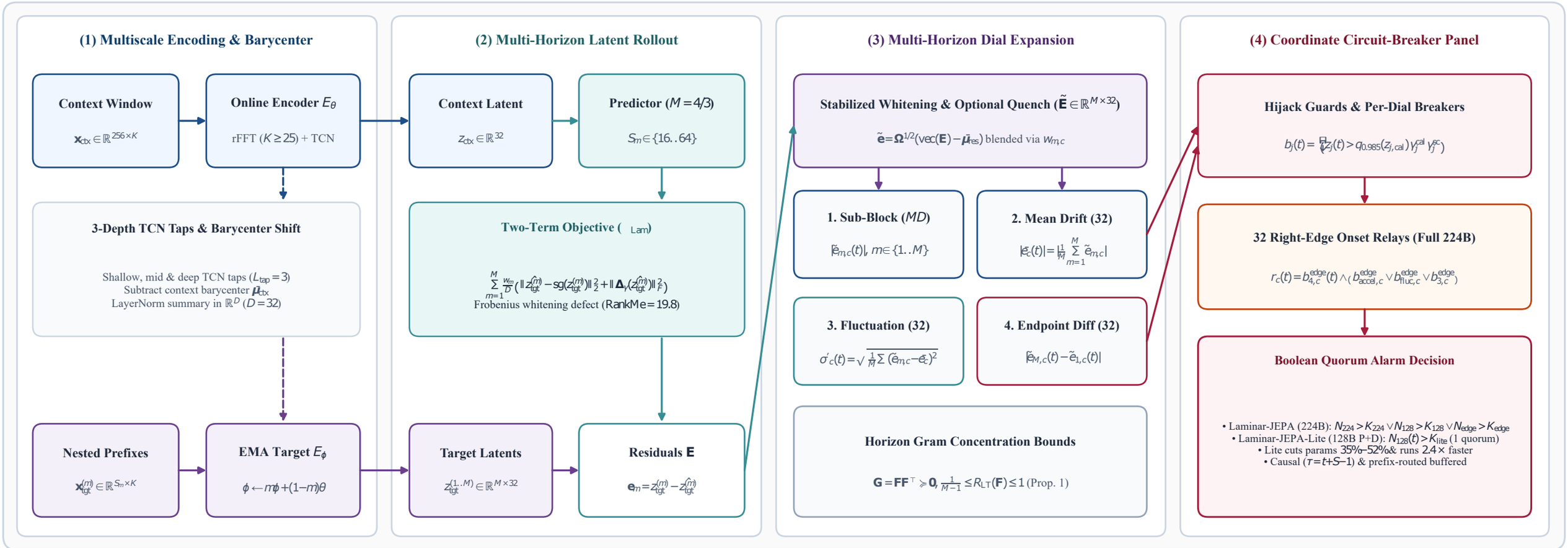

(1) Multiscale Encoding & Barycenter
Context Window
$\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$
Online Encoder E_θ
rFFT ($K \geq 25$) + TCN
3-Depth TCN Taps & Barycenter Shift
Shallow, mid & deep TCN taps ($L_{\text{tap}} = 3$)
Subtract context barycenter $\bar{\mu}_{\text{ctx}}$
LayerNorm summary in $\mathbb{R}^D$ ($D = 32$)
Nested Prefixes
$\mathbf{x}_{\text{tgt}}^{(m)} \in \mathbb{R}^{S_m\times K}$
EMA Target E_ϕ
$\phi \leftarrow m\phi + (1-m)\theta$
(2) Multi-Horizon Latent Rollout
Context Latent
$z_{\text{ctx}} \in \mathbb{R}^{32}$
Predictor ($M = 4/3$)
$S_m \in \{16..64\}$
Two-Term Objective (Lam)
$\sum_{m=1}^{M} \frac{w_m}{D}\left(\| z_{\text{tgt}}^{(m)} - \text{sg}(z_{\text{tgt}}^{(m)})\|_2^2 + \| \Delta_\gamma(z_{\text{tgt}}^{(m)})\|_F^2\right)$
Frobenius whitening defect (RankMe = 19.8)
Target Latents
$z_{\text{tgt}}^{(1..M)} \in \mathbb{R}^{M\times 32}$
Residuals $\mathbf{E}$
$\mathbf{e}_m = z_{\text{tgt}}^{(m)} - \hat{z}_{\text{tgt}}^{(m)}$
(3) Multi-Horizon Dial Expansion
Stabilized Whitening & Optional Quench ($\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$)
$\tilde{\mathbf{e}} = \Omega^{1/2}(\text{vec}(\mathbf{E}) - \bar{\mu}_{\text{res}})$ blended via $w_{m,c}$
1. Sub-Block (MD)
$|\tilde{e}_{m,c}(t)|,\ m \in \{1..M\}$
2. Mean Drift (32)
$|\bar{e}_c(t)| = |\frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}|$
3. Fluctuation (32)
$\sigma'_c(t) = \sqrt{\frac{1}{M}\sum(\tilde{e}_{m,c} - \bar{e}_c)^2}$
4. Endpoint Diff (32)
$|\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$
Horizon Gram Concentration Bounds
$\mathbf{G} = \mathbf{F}\mathbf{F}^\top \succcurlyeq \mathbf{0},\ \frac{1}{M-1} \leq R_{\text{LT}}(\mathbf{F}) \leq 1$ (Prop. 1)
(4) Coordinate Circuit-Breaker Panel
Hijack Guards & Per-Dial Breakers
$b_j(t) = \mathbb{1}(z_j(t) > q_{0.985}(z_{j,\text{cal}})\, v_j^{\text{cal}}\, v_j^{\text{sc}})$
32 Right-Edge Onset Relays (Full 224B)
$r_c(t) = b_{4,c}^{\text{edge}}(t) \wedge (b_{\text{accel},c}^{\text{edge}} \vee b_{\text{fluc},c}^{\text{edge}} \vee b_{3,c}^{\text{edge}})$
Boolean Quorum Alarm Decision
• Laminar-JEPA (224B): $N_{224} > K_{224} \vee N_{128} > K_{128} \vee N_{\text{edge}} > K_{\text{edge}}$
• Laminar-JEPA-Lite (128B P+D): $N_{128}(t) > K_{\text{lite}}$ (1 quorum)
• Lite cuts params 35%–52%& runs 2.4× faster
• Causal ($\tau = t+S-1$) & prefix-routed buffered